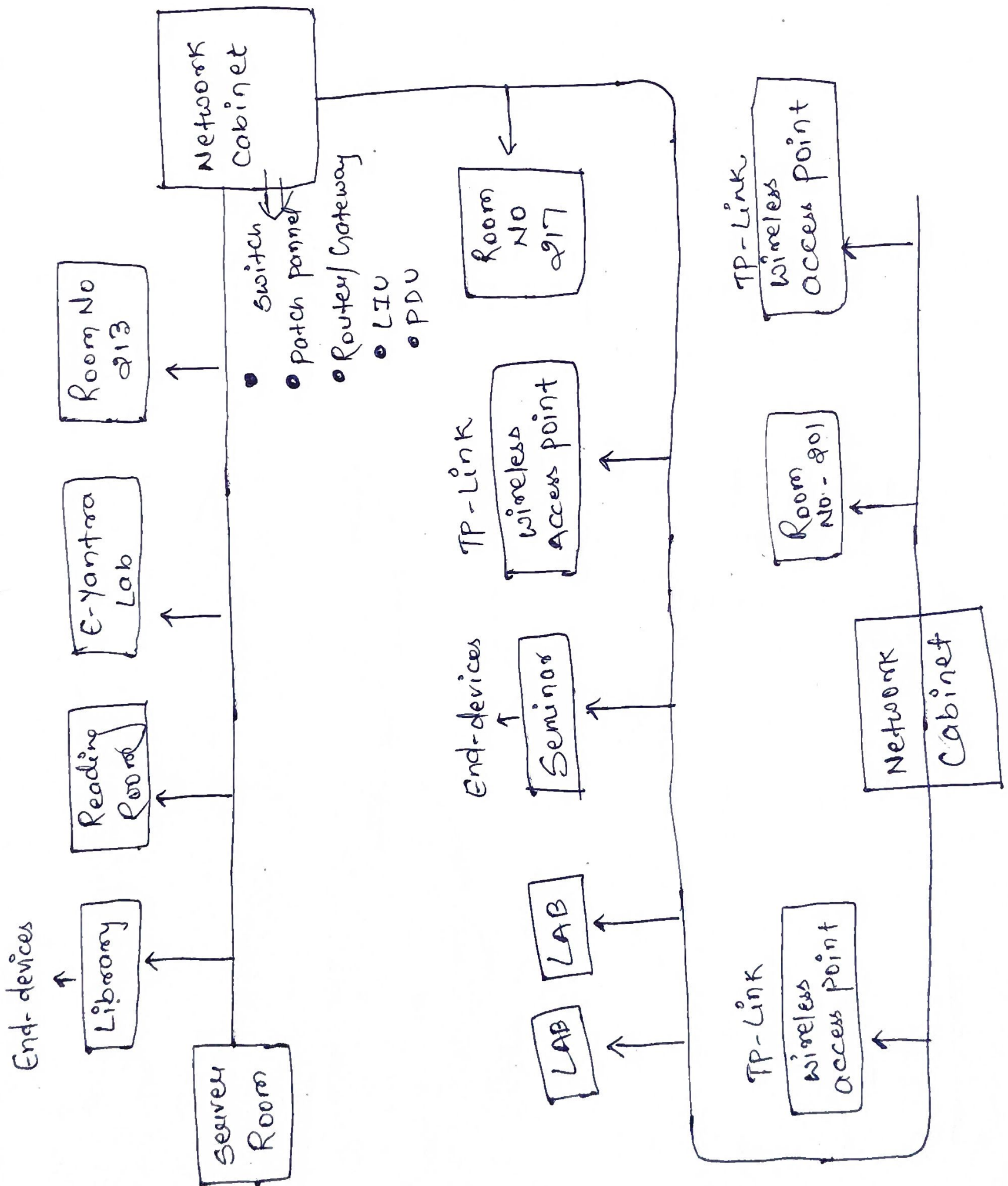


① Identify and Draw LAN Topology of your organization.

Clearly label all components such as Router, Switch, Access point and end devices.



2. Identify Network devices

a). List all networking devices used in LAN

b). Mention the function of each device.



→ (i) Network switch

→ It connects multiple devices (computers, printers, access points) within same local network.

→ It operates primarily at layer 2 (data-link layer) of the OSI model and uses MAC addresses to forward data directly to the specific destination device.

(ii) Router :-

→ Connects different networks together (such as connecting your LAN to the internet or connecting multiple subnetworks).

(iii) wireless access point (WAP)

→ Connects wireless devices (smartphones, laptop, tablets) to a wired network using Wi-Fi radio signals.

(iv) Network interface card (NIC)

→ The hardware component installed in a device (or integrated into its motherboard) that allows it to physically connect to a network via an Ethernet cable or wirelessly via wifi.

(v). Patch panel

→ A passive networking component used to organise, manage and terminate incoming and outgoing Ethernet cables in a central location.

(vi) Modem (Modulator - Demodulator)

→ Converts signals from an internet service provider (ISP) - such as fiber, coaxial cable, or DSL - into digital data that local network devices (like a router) can understand.

(vii) Firewall

- Monitors and filters incoming and outgoing network traffic based on predefined security rules.
- It acts as a security barrier b/w the trusted internal LAN and untrusted external network.

(viii) Network Hub

- An older, simpler device that connects multiple computers in a network. Unlike a switch, it broadcasts all incoming data packets to every connected port which can cause network congestion and security issue.

③ Analyze Connection Type :-

a). specify whether connections are :-

→ Wired (Ethernet) → for, classroom Smartboard, Library Computers.

→ Wireless :- for Students Smartphone & Laptop.

Wireless LAN adapter Wi-Fi: ipconfig /all

```
Connection-specific DNS Suffix . :  
Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz  
Physical Address. . . . . : FC-6D-77-65-F6-17  
DHCP Enabled. . . . . : Yes  
Autoconfiguration Enabled . . . . : Yes  
Link-local IPv6 Address . . . . . : fe80::451e:4f7b:9141:9b93%17(Preferred)  
IPv4 Address. . . . . : 10.85.23.254(Preferred)  
Subnet Mask . . . . . : 255.255.248.0  
Lease Obtained. . . . . : 21 July 2026 09:29:58  
Lease Expires . . . . . : 22 July 2026 11:17:06  
Default Gateway . . . . . : 10.85.16.1  
DHCP Server . . . . . : 10.85.16.1  
DHCPv6 IAID . . . . . : 166244549  
DHCPv6 Client DUID. . . . . : 00-01-01-00-30-FA-DB-0C-FC-6D-77-65-F6-17  
DNS Servers . . . . . : 8.8.8.8  
                        1.1.1.1  
NetBIOS over Tcpi . . . . . : Enabled
```

PS C:\Users\Ankush Raj> **Get-NetAdapter**

Name	InterfaceDescription	ifIndex	Status	MacAddress	LinkSpeed
Bluetooth Network Conn...	Bluetooth Device (Personal Area Netw...	18	Disconnected	FC-6D-77-65-F6-1B	3 Mbps
Wi-Fi	Intel(R) Wi-Fi 6 AX201 160MHz	17	Up	FC-6D-77-65-F6-17	130 Mbps
VMware Network Adapte...	VMware Virtual Ethernet Adapter for ...	16	Up	00-50-56-C0-00-08	100 Mbps
Local Area Connection	OpenVPN Data Channel Offload	40	Disconnected		1 Gbps
VMware Network Adapte...	VMware Virtual Ethernet Adapter for ...	13	Up	00-50-56-C0-00-01	100 Mbps
UrbanVPN	UrbanVPN Tunnel	39	Disconnected		100 Gbps

PS C:\Users\Ankush Raj> |

⑥ Briefly explain

a) Difference b/w wired and wireless speed

feature	wired speed (Ethernet)	wireless speed (Wi-Fi)
Max-real world speed	Consistently High	Variable
Latency (ping)	Ultra-low	Higher and Fluctuating
Stability/ Packet loss	Near 0% packet loss	Subject to packet loss from physical obstacles and frequency Congestion.
Band width Distribution	full dedicated bandwidth per physical Cable Connection.	Shared total band width among all wirelessly Connected devices.
Distance impact	Constant speed upto 100m	Speed drops significantly as a distance from AP/Router increases.

⑥ Role of Subnet mask in networking.

- Differentiating Network ID from Host ID
- Determine Local vs Remote Traffic
- Reduce Broadcast Congestion.
- Enables Network Segmentation, and security
- Optimizes IP Address Allocation.